

Jianbin Chen, CFA

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Summary

Focused on leveraging Machine Learning techniques in Asset Pricing, with expertise in Empirical Finance and predictive modeling. Experienced in developing data-driven methodologies for financial analysis, investment research, and Causal Inference.

Education

2021 - Present	Ph.D. Economics	University of California, Riverside	CA, US
2023 - Present	M.S. Computer Science	University of California, Riverside	CA, US
2021 - 2023	M.A. Economics	University of California, Riverside	CA, US
2008 - 2012	B.S. Mathematics	Shenzhen University	China

Certifications

CFA® Charterholder , CFA Institute	2018
CICPA (The Chinese Institute of Certified Public Accountant)	2018
Chinese Legal Professional Qualification Certificate	2016

Work Experience

Guosen Securities Company Limited <i>Senior Associate, Investment Banking Department</i> - Independently authored sections on business and financial analysis for Initial Public Offering (IPO) prospectuses. - Performed comprehensive commercial due diligence for significant assets. - Applied various valuation models to determine the fair value of subsidiaries. - Negotiated with prospective buyers and formulated transaction details for acquisition deals.	Shenzhen, China <i>Dec 2017 - Apr 2019</i>
Dongxing Securities Company Limited <i>Senior Associate, Investment Banking Department</i> - Assisted companies in aligning their accounting practices and business processes with SEC requirements. - Reviewed and analyzed financial statements, meticulously scrutinizing the reconciliation of accounting entries.	Shenzhen, China <i>Feb 2017 - Dec 2017</i>
PricewaterhouseCoopers (PwC) <i>Senior Associate, Assurance</i> - Utilized SQL to extract and consolidate business and financial data from over a thousand subsidiaries, and employed VBA and Python for advanced data analysis and reporting. - Managed assurance projects for publicly listed companies and multinational corporations (MNCs). - Conducted rigorous audit procedures across various accounts, including revenue, expenditure, debt, and receivables. - Executed asset impairment tests and constructed earnings forecasts using diverse models. - Verified the accuracy of accounting entries, identifying potential instances of financial fraud.	Shenzhen, China <i>Oct 2014 - Feb 2017</i>

PROJECTS

Advanced Algorithms Proficiency Demonstrated expertise in the implementation and analysis of advanced algorithms across various computational methodologies. Include: Randomized Algorithms, High Probability Analysis, Max-Flow Algorithms, Approximate Algorithms, Bitmask DP, Nearest Neighbor Search, Computational Geometry.
Parallel Algorithms - Designed and implemented race-free, wait-free, and lock-free algorithms utilizing Compare-and-Swap operations. - Developed a parallel quicksort algorithm, significantly enhancing runtime efficiency. Achieved a reduction in execution time from 6 seconds to 1.5 seconds on a 48-thread computer.

Deep Learning

- **CNN:** Employed Convolutional Neural Networks for the identification of items in the CIFAR-10 dataset.
- **VAE:** Trained a Variational Autoencoder on spiral data and generated 500 new samples for visualization.
- **LLM:** Utilized Microsoft phi-2 model to generate strings for a Product of Experts (PoE) model, calculating the conditional negative log likelihood of the outputs. Compared the performance of prompt concatenation to PoE using various samples, including random and hand-crafted prompts.

Breast Cancer Diagnostic Analysis

- **Description:** Conducted a comprehensive analysis on the Wisconsin Breast Cancer Diagnostic dataset, implementing a variety of machine learning techniques from scratch to improve diagnostic accuracy and data patterns.
- **Supervised Learning Techniques:**
 - Decision Tree & Naive Bayes Classifiers: Implemented with Information Gain splitting and Gaussian modeling; evaluated with stratified 10-fold cross-validation and F1 score.
 - Dimensionality Reduction: Applied SVD on training data, analyzing performance by SVD rank.
 - Feature Selection: Used randomization to assess feature predictiveness and enhance interpretability.
 - Data Augmentation: Balanced dataset with SMOTE, evaluating performance with oversampling percentages.
- **Unsupervised Learning Techniques:**
 - k-means: Implemented Lloyd's algorithm and k-means++; compared k values using Silhouette coefficient.
 - DBSCAN: Developed DBSCAN, evaluating with MinPts and Eps values.
 - Spectral Clustering: Implemented Unnormalized Spectral Clustering with Gaussian similarity.
 - Anomaly Detection: Detected anomalies with Isolation Forest, analyzed impact on k-means clustering.

Research & Publications

- **Jianbin Chen, Ke Peng.** [Stock Price Effect of CEO Turnover and Its Contributing Factors.](#) *South China Finance, 2014(07). P60-68*
- Predicting Earnings Change Directions Using a Feedforward Neural Network. *Work in process*
- Enhancing Fama-French Factor Models: Steps Towards Improvement. *Work in process*

Skills & Language

Skill: Financial Analysis, Quantitative Analysis, Econometric Modeling, Causal Inference
Tools: MS Office, Python, Pytorch, Numpy, Pandas, MySQL, L^AT_EX, C++(Basic)
Language: Mandarin(Native), Cantonese(Native), English(Fluent)